

Does Democratizing Search for Relevant Patent Information Facilitate Knowledge Recombination?

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Abstract

This study examines whether reducing search costs for relevant patent information increases corporate innovation. I exploit the 2013 adoption of the Cooperative Patent Classification (CPC) by U.S. and European patent authorities as a natural experiment. The CPC harmonizes patent classification across jurisdictions without altering the underlying disclosures, thereby reducing the cost of identifying relevant prior patents. Following the CPC adoption, public U.S. firms with greater exposure to non-English European patents experience larger increases in innovation activity relative to less exposed firms. This gain is driven by improved efficiency in knowledge recombination: firms generate more output per unit of cumulative R&D, rely on more related prior knowledge in developing innovations, and increase their use of European patents. The effects are stronger among firms facing greater pre-existing information barriers and are partially offset by European import competition. Overall, the findings highlight the importance of searchability in patent disclosure in facilitating innovation.

Keywords: Corporate Innovation, Research and Development, Search Costs, Patent Classification Harmonization

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1 Introduction

This study examines whether reducing search costs for relevant patent information facilitates innovation. Innovation arises through the recombination of existing knowledge, but effective recombination depends on firms' ability to navigate vast information sets and identify patent disclosure that are pertinent to their own innovation activities. The central question, therefore, is whether they can find the subset of information that is most relevant to their innovation activities. Lower search costs for relevant patent information can improve the efficiency of knowledge recombination and, in turn, foster greater innovative activity.

To investigate this question, I exploit the January 2013 adoption of the Cooperative Patent Classification (CPC) system as a natural experiment. The CPC harmonized fragmented classification systems in the U.S and Europe into a standardized, concept-driven, language-neutral framework. This harmonization makes it easier for U.S. firms to locate relevant European patents, especially those filed in French and German, the EPO's official languages alongside English. Importantly, the CPC changed only how disclosed patent documents were organized and classified without changing the timing, quality, language, or substantive content of patent disclosures. As a result, it provides a clean and plausibly exogenous shock to the search cost of relevant patent information.

I implement a Bartik instrument framework that exploits cross-sectional variation in firms' exposure to European patent disclosures. I construct firm-level treatment intensity based on a firm's predetermined exposure to non-English written and English written European patents interacted with a common CPC shock. This distinction between non-English and English is central, as non-English disclosures shall be less searchable to public U.S. firms prior to CPC. Firms with greater exposure to non-English patents therefore are expected to benefit more from the adoption of CPC. They constitute my treatment group. Firms with lower exposure serve as my control group. Comparing these groups before and after the CPC rollout allows me to estimate the effect of reducing search costs on corporate innovation.¹

¹This empirical specification can also be interpreted as a difference-in-difference design with continuous treatment, as a special case of a Bartik instrument (Breuer 2022)

The results show support for increased innovation resulting from reducing search costs for relevant European patents. Specifically, I find that a one standard deviation increase in firm-level exposure to non-English European patent disclosures is associated with approximately an 7.66% increase in the firm's innovation activity, measured by the number of U.S. patent citations received by patents filed in that firm year. In supplemental analyses, I also find that U.S. firms increase the quality of their innovation activity following the CPC adoption, measured by the average number of U.S. patent citations per patent.

Naturally, patent-based measures of innovation activity may be influenced by a firm's patenting strategy or the patent disclosure environment, raising concerns about whether the observed increase captures genuine innovation. To address this, I examine the real effects of reducing search costs for relevant patents. Specifically, I find that public U.S. firms with more exposed to non-English European patents increase their R&D spending, capital expenditures, and net inventor hiring after the adoption of CPC. These results suggest that the adoption of CPC leads to a real increase in innovation activity.

Next, I validate that the observed increase in innovation activity is due to lowered search cost of relevant patent information after the adoption of CPC. First. I find that public U.S. firms with greater exposure to non-English European patents firms generate more innovation output per unit of cumulative R&D, suggesting that U.S. firms are more efficient in knowledge recombination. Second, at the patent level, I find that new patents increasingly draw on U.S. and European knowledge that is more closely related to firms' existing knowledge base, suggesting U.S. firms engage in more focused knowledge recombination rather than a diffuse one. Third, I find that affected public U.S. firms increase citations to European patents after the CPC adoption, especially citations to non-English European patents, suggesting that they can more easily find relevant European patents. Together, these results suggest that the adoption of CPC facilitates innovation by lowering the search costs of relevant patent information and, in turn, improving the efficiency of knowledge recombination.

I then conduct cross-sectional tests to identify when reduced search frictions are most benefi-

cial and when their effects are attenuated by potential costs. First, I examine heterogeneity based on firms' ex ante information disadvantages. I find stronger innovation gains among U.S. firms without significant subsidiaries in France or Germany and among smaller U.S. firms. These results suggest that CPC adoption democratizes the search for relevant European patent information, disproportionately benefiting firms in information disadvantage. Second, I examine whether these benefits are moderated by potential costs. I find that U.S. firms in industries with lower exposure to European import competition experience stronger innovation gains following CPC adoption, indicating that the benefits of reduced search frictions are constrained by firms' competitive environment.

To expand the scope of the main findings, I extend the analysis to private U.S. firms and to the CPC subclass level. I find that private U.S. firms with greater exposure to non-English European patents increase their innovation activity following the CPC adoption. Importantly, similar effects are observed for U.S. firms with more exposure to English-language European patents, suggesting that the underlying search friction extends beyond language alone and reflects broader information frictions. In the CPC subclass level test, I find technology subclasses with greater ex ante exposure to non-English European patents experience larger increases in U.S. patent filings after the reform, indicating that the benefits of reduced search costs for relevant patent information extend beyond individual firms.

One potential concern is that improvements in machine translation technologies in the 2010s, such as the 2011 cooperation between Google Translate and EPO, may confound the observed results (Büttner et al. 2022). However, this explanation is unlikely to fully account for my findings. While such translation tools may reduce the costs of interpreting European patents, they do not help users search for relevant patents. To further address this concern, I use a quarterly sample and restrict the pre-period to quarters after the introduction of this cooperation. The results remain consistent. Finally, private firm evidence further weakens a pure translation explanation, as innovation also increases for private firms with more exposure to English-language European patents.

This paper makes three main contributions. First, it contributes to the accounting and innovation literature by showing that the effects of patent disclosure depend not only on what is disclosed

and when it is disclosed, but also on how easily the disclosed information can be searched. Prior research has mainly focused on the timing and content of patent disclosure and their effects on subsequent innovation (Kim et al. 2021; Dyer et al. 2024). Much less is known about whether search frictions limit the usefulness of disclosed patent information and, in turn, shape firms' innovation outcomes (as discussed in Glaeser et al. 2023). Here, I provide direct evidence that the searchability of patent documents plays an important role in shaping innovation outcomes.

Second, this study contributes to the literature on the real effects of information dissemination on corporate innovation. Prior work has largely focused on how the dissemination of financial disclosures influences R&D, showing both positive effects on follow-on innovation and efficiency (Chawla 2023), as well as unintended consequences such as reduced investment in innovation (Dambra et al. 2024; Chang et al. 2024). I extend this literature stream by focusing on patent information rather than financial disclosure. This distinction is important because patent disclosures contain detailed technical knowledge about existing inventions and therefore directly shape future R&D efforts.

Finally, this paper is also related to the literature that studies the design of patent information infrastructure. Prior research has examined how various characteristics of the patent system, such as patent disclosure timeliness (Kim et al. 2021; Hegde et al. 2023), patent examiners' leniency (Dyer et al. 2024; Shu et al. 2022), and patent grant rule (Valentine et al. 2025), influence corporate action and/or innovation outcomes. I add to this literature by focusing on patent classification, showing that information architecture is a key part of patent system design in shaping future innovation.

2 Institutional Details

2.1 Patent Classification System

A central goal of the patent system is to disseminate knowledge through patent disclosures and, in doing so, stimulate subsequent innovation by other inventors. Prior research highlights the importance of this disclosure function in shaping follow-on innovation, deterring product-market

competition, reducing information asymmetry, and generating unintended costs such as talent loss (Kim et al. 2021; Glaeser et al. 2021; Beyhaghi et al. 2022; Glaeser et al. 2025).

The effectiveness of the patent system’s disclosure function depends not only on making patent information public, but also on making it searchable. The searchability of patent information is particularly important given the sheer volume of patent documents. In 2023 alone, approximately 3.6 million patent applications were filed worldwide (*Patents Highlights* n.d.). Without a structured system to organize this information, inventors, firms, regulators, and examiners would face substantial difficulty finding relevant prior art and related knowledge.

Patent classification systems are designed to address this problem. For each disclosed invention, patent examiners assign classification codes that indicate the technological areas to which the invention belongs. These codes serve as a standardized indexing system for patent information. Much like keywords in academic research, they help users identify relevant documents more efficiently.

The importance of patent classification systems for information search has long been recognized in practice. Patent offices rely on classification codes to route applications to examiners with relevant technical expertise and to facilitate prior-art searches during examination (Dyer et al. 2024). Patent databases also rely on these classification codes to search for relevant patents; for example, Google Patents supports classification-based search. Researchers likewise use patent classifications to measure technological relationships and characterize innovation outcomes, such as patent originality, green innovation, and technological distance (Chawla 2023; Cheng et al. 2025; Akcigit et al. 2016; Zhang 2024). Taken together, these uses suggest that patent classification is a core part of the information infrastructure that shapes how easily relevant knowledge can be discovered and used.

2.2 Cooperative Patent Classification (CPC)

Prior to the implementation of CPC, the U.S. primarily relied on the United States Patent Classification (USPC) system, while Europe used the European Classification (ECLA) system. As

a result, even when patent information is publicly available, users faced substantial difficulty in identifying relevant prior art across jurisdictions.²

The Cooperative Patent Classification (CPC) system was introduced to address this problem. Implemented in January 2013, the CPC applies a standardized, concept-driven, and language-neutral classification scheme to USPTO and EPO patent documents. By replacing fragmented and office-specific classification practices with a shared system, the CPC made it easier for users to search for relevant patent information across borders.

The adoption of CPC is well-suited for empirical analysis for several reasons. First, it changes how patent authorities organize and classify patent documents without changing the timing, quality, language, or substantive content of patent disclosures. This makes it a clean shock that lowers the search costs by enabling inventors to search for relevant patents through a shared classification system.³ Second, exposure to this harmonization shock varies significantly across U.S. firms due to differences in their exposure to European patent information. This heterogeneity allows for the identification of causal effects using a continuous difference-in-differences approach, as a special case of Bartik instrument. Finally, the adoption of the CPC is plausibly exogenous from the perspective of U.S. firms: it is unlikely that firms preemptively adjusted their technological focus in anticipation of a classification system change. Together, these factors enable a clear examination of the causal effect of reductions in search frictions on corporate innovation.⁴

²Although both offices assigned International Patent Classification (IPC) codes to patent documents for international consistency, the IPC was less granular, updated less frequently, and—most importantly—was not the primary classification system used by either authority. Instead, each office relied on its own classification scheme to allocate patent applications to examiners and search for prior art. As a result, classification practices and interpretations of IPC diverged across jurisdictions, creating barriers for users seeking to identify relevant prior art.

³See **Figure 1** and **Figure 2** for examples

⁴A potential concern is that contemporaneous changes to the U.S. patent system, such as the adoption of the Leahy–Smith America Invents Act (AIA) in September 2012, may confound the results. However, the AIA primarily altered patent filing incentives by transitioning the U.S. system from a “first-to-invent” to a “first-to-file” regime, without affecting the searchability of patent information (Valentine et al. 2025). Moreover, because the AIA applied uniformly across firms, it is unlikely to explain the variation in outcomes based on firms’ exposure to European patents.

2.3 Distinguishing Cooperative Patent Classification (CPC) from EDGAR and American Inventors Protection Act (AIPA)

Prior research has primarily examined the role of financial disclosure dissemination in shaping corporate innovation, often using the implementation of the EDGAR system as a natural experiment (Chawla 2023; Dambra et al. 2024; Chang et al. 2024). EDGAR made financial disclosures electronically accessible, thereby reducing the costs of accessing and acquiring financial information. Importantly, in the financial disclosure setting, users know ex-ante what information they are looking for. By contrast, the patent setting involves a different type of information problem. Innovators typically do not know in advance which prior patents are most relevant for their research, and must instead search across a large and complex space of existing disclosures, much like researchers conducting literature review.

Existing studies of patent disclosure have studied on the enactment of American Inventors Protection Act (AIPA) on innovation (Kim et al. 2021; Hegde et al. 2023). The AIPA accelerated the USPTO's release of patent information by mandating the disclosure of U.S. patent applications 18 months after filing, regardless of whether the application is eventually granted. This setting inherently bundles changes in disclosure timing and the searchability of disclosures, making it difficult to disentangle the effect of earlier disclosure from the effect of reduced search costs for relevant patent information (Glaeser et al. 2023).

My study complements these studies by isolating the effect of lower search costs for relevant patent information on corporate innovation. The adoption of CPC can be viewed as complementary to the AIPA in facilitating knowledge flows. While the AIPA increases the availability of patent information by accelerating disclosure, the CPC increases the usability of that information by making it easier for inventors to find relevant patents within a large stock of existing disclosures. More broadly, my study highlights that the innovative value of patent disclosure depends not only on whether information is released, but also on how easily it can be searched.

3 Hypothesis Development

Reducing search frictions for relevant European patents can facilitate innovation by improving the efficiency with which firms recombine existing knowledge. New innovations often arise from the recombination of existing ideas. However, not all combinations are productive. Promising innovation requires linking ideas that are meaningfully related rather than arbitrarily combined, as not many people like chicken flavored ice cream (Jones 2023; Weitzman 1998). This process critically depends on firms' ability to identify relevant prior knowledge.

In practice, search frictions impede this process. Prior research documents that innovators exhibit local biases in geography and language when using patent information (Murata et al. 2014; Jaffe et al. 1993), suggesting that they disproportionately rely on familiar and easily accessible knowledge. As a result, firms may waste resources seeking unproductive directions, overlook promising opportunities, and unnecessarily reinvent the wheel when they fail to identify relevant prior art (Krieger 2021; Capkun et al. 2023; Zhang 2023; Hegde et al. 2023). By improving the organization and indexing of patent information, the CPC helps firms identify relevant European patent disclosures, enabling a more focused search process that improves the efficiency of knowledge recombination, and, in turn, fosters greater innovation.

For public U.S. firms, these search frictions are especially salient for non-English European patents because established public firms may already have the capacity to identify relevant English-language European patent information. Accordingly, the main empirical analysis focuses on public U.S. firms' exposure to non-English European patents. My main hypothesis, stated in alternative form, is as follows:

Hypothesis 1. After the adoption of the CPC, U.S. firms with greater exposure to non-English European patent disclosures exhibit a larger increase in innovation activity compared to those with less exposure to non-English European patent disclosures.

However, the effect of CPC adoption is not without tension. First, search frictions may not have been severe enough to materially constrain innovation for all firms. In particular, large firms or

those with operations in France or Germany may have already overcome cross-border information barriers using existing resources, local expertise, or internal search capacity.

Second, the reduction in search costs resulting from the adoption of CPC is a double-edged sword. While U.S. firms can more easily identify relevant European knowledge, European firms can likewise find U.S. patents more readily, potentially intensifying competition. This increased transparency may facilitate rival innovation and talent reallocation, which can dampen firms' incentives to innovate (Glaeser et al. 2025; Kim et al. 2021; Chang et al. 2024; Dambra et al. 2024). Therefore, the overall effect of the adoption of CPC on innovation remains an empirical question.

4 Data and Variable measurement

4.1 Dependent Variable: Innovation measures

Similarly to the prior literature, I use the number of US patent-generated citations a firm's patents receives over the patents' lifetime as a proxy for the value of patents (Kim et al. 2021; Bernstein 2015; Zhong 2018). I count only US patent-generated citations to avoid confounding my measurement with the increase in European counterparts' ability to cite US patents after the adoption of CPC. Further, I measure forward citations through 2024 instead of measuring through some constant years because there is an endogenous time lag between when firms file their patents and when the patent applications are available for the public to cite, as firms may choose to disclose their patent documents earlier to deterring product market competition (Glaeser et al. 2021).

Given my focus on firm-year-level innovation activity, I aggregate US patent-generated forward citations a firm receives for all patents filed in a given firm-filing year, which I refer to as *Forward Citations*. I use *Forward Citation* with an inverse hyperbolic sine transform as the primary proxy for innovation. I apply inverse hyperbolic sine to the outcome variables to account for their skewed distribution and minimize the impact of outliers in my sample (Glaeser et al. 2022).⁵

⁵As shown in **Table 13** and **Table 14**, the results are qualitatively similar when I use *Scaled Forward Citations* to address potential time truncation concerns (Lerner et al. 2022) or estimate the model using fixed-effect Poisson regressions.

In addition, I examine the real effect of the adoption of CPC by focusing on firms' R&D investment, capital expenditures, and net inventor hiring. These tests help rule out the possibility that the main results are driven by changes in patenting strategy or the patent disclosure environment rather than real changes in innovative activity.⁶

4.2 Independent Variable: Firm-level Exposure to the CPC Adoption

My analysis involves quantifying U.S. firms' exposure to both English and non-English European patent disclosures to proxy for the impact of patent classification harmonization. My approach measures a firm's exposure to the CPC adoption by examining its level of exposure to English and non-English European patent disclosures within the technology fields where it conducts R&D (Mezzanotti 2021; Breuer 2022). These technology fields are categorized by the CPC. My method capitalizes on two key features of the setting. First, there is a lot of variation across the technology fields in the prevalence of European patent disclosure. Second, many companies operate across different technology fields.

I express an individual firm i exposure to the adoption of CPC as a function of two quantities. (1) the technology fields j in which the firm i operates, and (2) the prevalence of European patent disclosure across the technology fields j . Specifically, I define σ_j^i as the share of firm i patents across the different technology fields j and $\rho_{l,j}$ as the prevalence of European patent disclosure in technology fields j , where l indicates language (English vs. Non-English). Therefore, firm i exposure to English or non-English European patent disclosure can be constructed by weighting the prevalence of European patent disclosure in each technology field by the share of activity that firm i has in each of these fields, which is:

$$Exposure_{i,l} = \sum_{j=1}^T \sigma_j^i \rho_{l,j}$$

To proxy for a firm i exposure in each technology field j , σ_j^i , I compute the ratio of US patents filed by firm i within each technology field j from 1993 to 2012 to the total number of US patents

⁶Firms might strategically split inventions into multiple patents or cite more extensively post-CPC, inflating citation counts without a proportional rise in true innovation

filed by firm i during the same period. Here, the technology fields are defined by the CPC, which classifies patents into over 600 distinct subclasses.⁷

To estimate the prevalence of European patent disclosure across technology fields j , I use patent data from PATSTAT, a comprehensive patent database provided by EPO. PATSTAT provides detailed information on the language in which these European patents are written, enabling me to categorize them into English-written and non-English-written patents (Patents that are written in French and German – the other two official languages of EPO). Using patent filing data from 1993 to 2012, I calculate the prevalence of European patent disclosure across technology subclass j by dividing the number of English/non-English European patents by the number of US patents within each specific technology subclass. I present the prevalence of non-English European patent disclosure across technology sections (not subclasses) in **Figure 3** for demonstration.

I estimate $Exposure_{i,l}$ by combining the two measurements I mentioned above. I separately estimate two kinds of exposure based on the language written in the European patents, $Exposure_{En}$ and $Exposure_{Other}$.

I acknowledge that a firm may file multiple patent documents in different countries (Fink et al. 2016). Many patents filed in Europe are later filed in the United States. When computing firm-level exposure to the CPC adoption, I determine whether a patent is a U.S. or European patent based on the location of its initial filing. This approach assumes that the initial filing location reflects the patent’s origin and where its disclosure is first made available. In the robustness check, I find that my result is consistent and even stronger if I define a European Patent as a patent that is only filed in Europe, not subsequently filed in the US.⁸

4.3 Control Variable

I add a set of firm-level controls that may influence a firm’s innovation activities in my regression based on prior literature (Kim et al. 2021). More specifically, to control for financial friction,

⁷I use the current CPC classification to ensure a consistent and harmonized mapping of technological fields over time.

⁸See **Table 14**

agency conflict, and investment opportunity that may influence a firm's innovation activities, I include *Age*, *Firm Size*, *Cash*, *Return on Assets*, *Book to Market*, *Number of Analysts*, and *Leverage* as a set of controls. Detailed definitions of all variables are in the Appendix A.

4.4 Sample Selection

Table 1 describes the sample selection. My sample includes publicly traded companies from 2009 to 2016. I choose this time frame to ensure a balanced representation with four years in the pre-period and four years in the post-period. My sample period starts in 2009 to mitigate any potential influence from the 2008 financial crisis on the observed outcomes.

To compile the data set, I integrated financial statement data sourced from Compustat and analyst data from IBES. Additionally, I incorporated patent data initially obtained from Kogan et al. (2017) and supplemented it with more detailed patent data from PATSTAT and PatentsView.

I retain only firms that have non-missing values of my independent variable of interest. This criterion requires that a firm has filed a granted patent at least once in the twenty years leading to the adoption of the CPC. I also require firms to have relevant CRSP/Compustat variables, such as market capitalization, book assets, and returns.

Table 2 presents descriptive statistics. I report summary statistics for all variables in their raw form, prior to any transformations. As shown in **Table 2**, firms in the sample receive, on average, 189 forward citations to patents filed in a given year. The average firm age in the sample is approximately 23 years.

5 Research Design

To investigate whether reducing search costs for relevant patent facilitate corporate innovation, I estimate the following regression equation:

$$Innovation_{i,t} = \beta_1 Exposure_Other \times Post + \beta_2 Exposure_En \times Post + \beta_3 Controls_{i,t} \\ + \delta_i + \pi_t + \mu_{i,t}$$

The unit of observation is at the firm i , year t level. The dependent variable, $Innovation_{i,t}$, is one of several proxies used to capture firm-level innovation activity. The independent variable $Exposure_Other$ is the amount of exposure to non-English European patents. The independent variable $Exposure_En$ is the amount of exposure to English written European patents. Both of them are absorbed by the firm fixed effect. The coefficients on the interaction terms $Exposure_Other \times Post$ and $Exposure_En \times Post$ capture the impact of the CPC. By Hypothesis 1, I predict $\beta_1 > 0$. I have no directional prediction for β_2 because public U.S. firms may already be familiar with English-written European patents. Therefore, the effect may be minimal. I define firm and year fixed effects as δ_i and π_t , respectively, and I cluster standard errors at the firm level.

6 Empirical Results

6.1 Innovation Activity

Table 3 regresses *Forward Citation* on $Exposure_Other \times Post$ and $Exposure_En \times Post$. Consistent with the hypothesis, the coefficient on $Exposure_Other \times Post$ is positive and statistically significant. In Column (2), the estimated coefficient (coef. = 1.16; t-stat = 2.74) implies that, following the adoption of CPC, a one standard deviation increase in firm-level exposure to non-English European patent disclosures is associated with an approximately 7.66% increase in forward citations relative to the pre-CPC average. This suggests that improvements in the organization and indexing of patent information alone can lead to a substantial increase in innovation activity by reducing search costs for relevant knowledge. The coefficient on $Exposure_En \times Post$ is not statistically significant, which is consistent with the notion that public U.S. firms may already be familiar with English-written European patents and therefore do not benefit from the adoption of CPC.

To further examine the nature of the effect, I analyze *Forward Citation per Patent* in Columns (3) and (4) of **Table 3**. The positive and significant coefficients on $Exposure_Other \times Post$ indicate that, following the CPC adoption, treated firms not only increase the quantity of their innovation activity but also increase the quality of their innovation.

A key identification assumption for my main tests is parallel trends. To provide suggestive evidence, I plot the coefficients and their 90% confidence intervals around the patent classification harmonization, using the model below:

$$Innovation_{i,t} = \beta_1 Exposure_Other \times Year_t + \beta_2 Exposure_En \times Year_t + \beta_3 Controls_{i,t} + \delta_i + \pi_t + \mu_{i,t}$$

The unit of observation is at the firm i , year t level. The dependent variable, $Innovation_{i,t}$, is *Forward Citation*. $Exposure_Other$ is the amount of exposure to non-English European patents. $Exposure_En$ is the amount of exposure to English written European patents. Both of them are absorbed by the firm fixed effect. The coefficients on the interaction terms $Exposure_Other \times Year_t$ and $Exposure_En \times Year_t$ capture the dynamic impact of the CPC. I define the firm and year fixed effects as δ_i and π_t , and I cluster standard errors at the firm level. **Figure 4** shows that the coefficient estimates for the pre-event years are virtually indifferent from zero, hence validating the parallel trends assumption.

6.2 Real Effect

One concern of previous results is that the observed increase in innovation may be driven by changes in patenting strategy or the patent disclosure environment rather than a genuine increase in innovation activity. To address this concern, I examine the real effects of reduced search costs for relevant patent information. Because these measures are less directly affected by patenting strategy and environment, they help rule out the possibility that the main results are driven by mechanical changes in patent-based outcomes.

Specifically, I analyze three proxies that capture the real effect of CPC: *R&D intensity*, *Capx*,

and *Net hire*. *R&D intensity* captures the proportion of firm resources directly allocated to developing new innovations. *CapEx* proxies investments in physical capital that support innovation, such as infrastructure and equipment required to develop, implement, and scale new technologies. Finally, *Net hire* measures firms' investment in innovation-related human capital, focusing on the net hiring of patent inventors.⁹

As shown in **Table 4**, the coefficients on *R&D intensity*, *CapEx*, and *Net hire* are all positive and statistically significant. These findings suggest that the reduction in search costs for relevant patent information is associated with real changes in firms' actions, providing complementary evidence that the adoption of CPC facilitate overall innovation activity.

6.3 Mechanism

6.3.1 Efficient Knowledge Recombination

I examine whether the reduction in search costs for relevant European patent information improves firms' efficiency in recombining existing knowledge. If the adoption of the CPC facilitates innovation by improving the efficiency with which firms recombine existing knowledge, affected firms generate more output per unit of cumulative R&D.

To test this, I construct a measure of innovative efficiency similar to the approach outlined in the previous literature (Hirshleifer et al. 2013; Chawla 2023). I define *Innovative Efficiency* as *Forward Citation* scaled by cumulative R&D capital. I calculate a firm's cumulative R&D capital as the firm's cumulative R&D expense over the last five years, with a depreciation rate of 20%.

Table 5 reports the results. In Column (2), the coefficient of *Exposure_{Other} × Post* is positive and statistically significant (coef. = 0.57; t-stat = 4.04), indicating that the adoption of CPC increases firms' innovative efficiency. This evidence is consistent with the proposed mechanism

⁹To construct *Net hire*, I use patent data from (Kogan et al. 2017) and inventor-level information from PatentsView, which provides unique inventor identifiers. Following prior studies (Mkrtchyan et al. 2024; Hombert et al. 2017; Li et al. 2022), I infer job transitions based on the midpoint between application years of consecutive patents filed with different firms. For example, if an inventor files a patent with Firm A in 2010 and another with Firm B in 2014, the transition is assumed to occur in 2012. The inventor is then classified as a new hire for Firm B and a departure from Firm A in that year.

that reducing search frictions for relevant patent information improves the efficiency of knowledge recombination, thereby facilitate innovation.

6.3.2 Focused Knowledge Recombination

To further solidify that more efficient knowledge recombination drives the increase in innovation, I conduct a patent-level analysis of the technological distance between the U.S. and European knowledge cited by new patents and firms' existing knowledge base. If the CPC facilitates corporate innovation by reducing search costs for relevant patents, then firms should engage in a more focused recombination of knowledge.

To implement this test, I construct a measure of *Technology Distance*, which captures how closely the U.S. and European patents cited in a firm's patent application align with the firm's existing technology base, following prior literature (Zhang 2024; Akcigit et al. 2016).¹⁰ A decrease in *Technology Distance* implies that firms are building new patents on knowledge that is more closely related to their existing knowledge base. To account for technology- and time-specific trends, I include CPC subclass-by-filing-year fixed effects in the regression.

Consistent with the proposed mechanism, as shown in **Table 6**, the coefficient in **Table 6** is negative and significant. This result implies that, following the adoption of CPC, affected firms' innovations increasingly rely on U.S. and European knowledge that is more closely aligned with their existing technological expertise. This finding suggests that public U.S. firms conduct more targeted and focused knowledge combinations, consistent with the improved knowledge recombination efficiency mechanism.

6.3.3 Knowledge Flow between Europe and the United States

One advantage of my setting is that I can directly measure knowledge flow using patent citations between Europe and the United States as a proxy. Prior research has established that information transfer is significantly more likely when a patent citation is observed than when it is

¹⁰The detailed measurement is in Appendix B.

not (Jaffe et al. 2000; Jaffe et al. 1993). If the adoption of the Cooperative Patent Classification (CPC) system enhances innovation through more efficient knowledge recombination by decreasing search costs for relevant European patents, then public firms operating in technology fields with greater exposure to non-English European patent disclosures should cite more European patents following the adoption of CPC.

To measure the extent of knowledge flow from Europe to U.S. firms, I calculate the total number of European patents cited—by either patent applicants or examiners—in patent applications filed by a firm in year t (EU). I also separately compute the number of European patents cited exclusively by the applicants in those filings ($EU\ Appl$). The $EU\ Appl$ proxy may underestimate applicant-driven citations because the database classifies examiner-cited references as “cited by examiner,” even if the applicant also cites them. Additionally, applicants may strategically omit prior art to improve grant chances (Langinier et al. 2016), while examiners, driven by productivity incentives, may not rigorously detect such omissions (Langinier et al. 2020).

Despite these measurement limitations, the results presented in Columns (1) and (2) of **Table 7** show that the coefficient on $Exposure_Other \times Post$ is positive and statistically significant. These results indicate that U.S. firms increase their use of European patent knowledge following the CPC adoption, consistent with the idea that lower search costs enable firms to more effectively find and build on previously less accessible prior art.

To further examine the mechanism, I distinguish between European citations to different subsets of European patents based on their pre-existing searchability. I construct separate measures for citations to non-English European patents EU_En and English European patents EU_Other . If the CPC adoption primarily operates by reducing search costs for less accessible relevant knowledge, then the increase in citations should be concentrated in the latter group.

Columns (3) and (4) of **Table 7** support this prediction. The coefficient on $Exposure_Other \times Post$ is positive and statistically significant for citations to non-English European patents, while the coefficients on $Exposure_En \times Post$ are small and not statistically significant. Overall, the evidence indicates that firms cite more European patents after the adoption of CPC, providing direct support

that the adoption of CPC facilitates corporate innovation by improving knowledge recombination efficiency.

6.4 Cross-sectional Tests

6.4.1 Firms' ex ante information disadvantages

This section examines how firms' ex ante information disadvantages shape the effect of CPC adoption on corporate innovation. The underlying idea is straightforward. If CPC adoption promotes innovation by reducing search costs for relevant European patent information, then its effect should be stronger for firms that faced greater barriers to accessing such information before the reform. I perform two cross-sectional analyses using proxies for firms' ex ante difficulty in finding relevant non-English European patent disclosures: subsidiary presence in France or Germany and the size of public firm.

The first proxy captures public firms' local presence in the main jurisdictions where many non-English European patent disclosures originate. I predict that the CPC adoption should have a stronger effect on U.S. firms without significant operations in France or Germany. These firms are less likely to possess local expertise, established patent-search networks, or familiarity with relevant European disclosures, and therefore likely faced higher search costs before the adoption of CPC. The second proxy is the size of public firm. I expect smaller public firms to benefit disproportionately from the CPC adoption because they are more likely to face resource constraints in searching for and processing European patent information relative to large firms. As a result, improvements in the organization and indexing of patent information should be especially valuable to them.

To measure U.S. firms' presence in France and Germany, I use Exhibit 21 of firms' 10-K filings, which lists significant subsidiaries as required by the SEC. Subsidiaries are disclosed if they meet any of the following thresholds: (1) investments or advances exceed 10% of consolidated assets, (2) assets exceed 10%, or (3) pretax income exceeds 10% of consolidated pretax income. Based on Exhibit 21 information, I construct an indicator variable, *No French and Germany Presence*, equal

to one if a firm had no significant subsidiaries in France or Germany in 2012 (pre-CPC adoption). To measure U.S. firms' size, I use total assets from Compustat and define *Small Firm* as an indicator equal to one if the firm is below the median asset size in 2012.

I then estimate cross-sectional specifications by interacting these firm characteristics with $Exposure_Other \times Post$ and $Exposure_En \times Post$, using *Forward Citations* as the dependent variable. **Table 8** reports the results, including the baseline specification and models with additional controls. Across Panels A and B, I find that the coefficients on *No France/Germany Presence* $\times Exposure_Other \times Post$ and *Small* $\times Exposure_Other \times Post$ are positive and statistically significant. These findings suggest that the positive effect of the CPC adoption on innovation activity is concentrated among firms with greater ex ante information disadvantages. It supports the view that the adoption of CPC promotes innovation by democratizing the search for relevant European patent information, and the benefits are stronger for firms that were most constrained by search frictions ex ante.

6.4.2 European Import Competition

In addition, I examine the case where the benefits of reduced search frictions may be attenuated by potential costs. Although the CPC decrease search costs for relevant European patent information for U.S. firms, it may also operate as a double-edged sword. The CPC may not only help U.S. firms search European knowledge more effectively, but also make U.S. patents more transparent and accessible to European competitors. This reciprocal increase in transparency can raise proprietary costs by increasing the risk of appropriation, intensifying product-market competition, and facilitating talent loss (Dambra et al. 2024; Glaeser et al. 2025; Kim et al. 2021). Therefore, the positive effect of the adoption of CPC on innovation shall be weaker in industries with greater exposure to European competition and stronger among firms facing lower European import penetration.

Following prior literature, I use European import penetration as a proxy for industry-level exposure to European competition (Lie et al. 2023). I calculate import penetration in the three-digit

SIC industry as:

$$Import\ Penetration = \frac{imports}{imports + domestic\ production - exports}$$

Import and Export data come from the United States Census Bureau, and domestic production data are obtained from the NBER-CES Manufacturing Industry Database. I construct an indicator variable, *Low Competition*, which equals one if a firm operates in an industry with below-median European import penetration in 2012.

To examine cross-sectional effects, I interact *Low Competition* with *Exposure_Other* $\times$ *Post* and *Exposure_En* $\times$ *Post*, using *Forward Citations* as the dependent variable. **Table 9** presents the baseline estimates along with the results including a set of control. Consistent with my prediction, the coefficient of *Low Competition* $\times$ *Exposure_Other* $\times$ *Post* is positive and significant. Combining the baseline and interaction coefficients implies that the marginal effect of CPC for U.S. firms in low-competition industries is approximately 4.05, compared to 1.50 in high-competition industries—more than twice as large. This finding suggests that the benefits of CPC for U.S. firms are substantially attenuated in more competitive environments. Overall, the evidence supports the view that the adoption of CPC operates as a double-edged sword: while it facilitates knowledge spillovers, its net effect on innovation depends on public U.S. firms' exposure to European competitive pressure. This finding is closely related to the proprietary cost channel emphasized by (Kim et al. 2021), which shows that timely patent disclosure can weaken innovation incentives.

6.5 Scope of Main finding

6.5.1 Private Firm

To expand the scope of the main finding, I conduct an additional test by analyzing the effect of the CPC adoption on private U.S. firms. This test is useful for two reasons. First, it helps assess whether the benefits of reduced search frictions extend beyond established public firms. Second, it provides further insight into the nature of the underlying search friction. If CPC adoption promotes innovation by reducing search costs for relevant European patent information, then its

effects should be at least as strong among private firms, which are likely to face greater barriers in searching European patent information.

To capture innovation activity more comprehensively, I examine three proxies: the number of patents, patent claims, and forward citations. Patent claims complement the other two measures because they capture the scope and breadth of the inventions firms produce. Thus, an increase in patent claims suggests that firms are producing broader and more substantial innovations.

Consistent with this prediction, in **Table 10**, I find that private U.S. firms with greater exposure to non-English European patents increase their innovation activity following the CPC adoption. The result is consistent with the evidence for public firms. More importantly, I also find a positive and significant effect for private firms with greater exposure to English written European patents. This finding suggests that search frictions extend beyond language barriers alone. Even when patents are written in English, private firms may still face substantial difficulties in searching relevant European patent information. By improving the classification of patent information, the adoption of CPC reduces these broader search frictions and facilitates private firms' innovation.

6.5.2 Aggregate Effect

To gauge the aggregate effect of the adoption of CPC on innovation, I extend the analysis to the technology-field level. Specifically, I aggregate innovation outcomes to the CPC subclass-year level and test whether technology fields with greater ex ante exposure to European patent information experience larger increases in patent filings by U.S. firms following the CPC adoption.

As shown in **Table 11**, I find that CPC subclasses with greater ex ante exposure to non-English European patents experience significantly higher increases in innovation activities after the adoption of CPC. Overall, the finding suggests that harmonizing patent classification at the cross-country level can have meaningful effects on the overall innovation landscape by reducing search frictions across jurisdictions.

6.6 Alternative mechanism: Google Patent Translation

A potential alternative explanation for the main results is that improvements in machine translation technologies, rather than the adoption of CPC, lead to the observed increase in corporate innovation (Büttner et al. 2022). Google Translate has supported translation into English from major European languages such as French and German since 2006 (Fang et al. 2025). Building on this, in March 2011, Google and the European Patent Office (EPO) launched a cooperation to enhance patent translation capabilities. This effort led to the introduction of Patent Translate in Feb 2012 as part of the EPO's Espacenet service, enabling translation of patent documents across multiple European and non-European languages (Google Translate Blog 2012). Therefore, it is possible that these improvements in machine translation technology partially explain the observed increase in innovation.

However, this explanation is unlikely to fully account for my findings. First, translation tools in the Espacenet service primarily help users interpret patent documents once they have been discovered. They do not directly address the challenge of searching for relevant patents within a large and complex information space.¹¹ In contrast, the CPC fundamentally improves the classification of disclosed patent information, thereby reducing the search costs associated with identifying relevant prior art, which leads to more efficient and focused knowledge recombination.

To further address this concern, I re-estimate the main specification using a quarterly sample and restrict the pre-period to quarters after the introduction of Patent Translate. This specification ensures that the availability of improved translation technology is held constant throughout the sample. As shown in **Table 12**, the results remain quantitatively and qualitatively similar to the baseline findings, suggesting that the observed effects are not solely driven by machine translation improvements.

Finally, prior evidence from the private firm analysis further weakens this interpretation. Specifically, I find that the adoption of CPC increases innovation not only for private firms with greater exposure to non-English European patents, but also for those with greater exposure to

¹¹ See European Patent Office 2012 for a video describing the outcome of Google–EPO cooperation.

English-language European patents. Since language barriers are absent in the latter case, this result suggests that CPC operates by improving the overall searchability of patent information, rather than merely alleviating language-related frictions.

6.7 Robustness Checks

6.7.1 Placebo Test

To alleviate the concern that my main findings may be spurious, I conduct a falsification test by randomly shuffling *Exposure_Other* to each firm and reestimating the regression equation. I repeat this process 500 times. **Figure 5** shows the distribution of the estimated coefficients and t-stats of *Forward Citation* on $Post \times Exposure_Other$ based on placebo regression using randomized *Exposure_Other*. The distributions for DiD estimates and the t-statics, based on placebo regressions, are centered around zero, which contrasts starkly with the estimated effects of **Table 3**. Therefore, my documented effect is unlikely to be spurious.

6.7.2 Alternative Innovation Proxies

I explore the robustness of my result to alternative dependent variable measurements. **Table 13** presents results. Column (1) shows that my results continue to hold using untransformed *Forward Citation*. Further, in Column (2), I apply the log plus one transformation to my independent variable and find that my result is consistent and robust. Column (3) uses *Scaled Forward Citations* to address potential time-truncation concerns (Lerner et al. 2022). The results remain consistent with the main findings. In addition, Columns (4) and (5) find that affected U.S. firms file more patents and are more likely to file any patents after the CPC adoption. In Column (6), I also find that firms increase their patent claims post-CPC adoption. Patent claims capture the scope and breadth of an invention. Therefore, an increase in patent claims suggests that firms are producing more substantial innovations. Finally, in Column (7), I use *Patent Value* to capture the private economic value of patents, measured by investor reactions on the patent grant date following (Kogan et al.

2017). The results remain robust.

6.7.3 Alternative Model Specification

In panel A of **Table 14**, I examine the robustness of my results to a variety of alternative *Exposure_Other* and *Exposure_En* proxies and fixed effect structure. One concern is that many patents filed in Europe are subsequently filed in the United States. To address this concern, I redefine European patents as those filed exclusively in Europe in my sample period and recalculate *Exposure_Other* and *Exposure_En*. Using these revised measures, I find that my results remain consistent and are even stronger, as shown in Column (1). In Column (2), I dichotomize treatment by defining *High Exposure_Other* and *High Exposure_En* as indicators equal to one if the non-English or English exposure of a firm is above the median and zero otherwise. The results remain robust, suggesting that the results are not driven by outliers or functional form. Further, in Column (3), I include firm fixed effects and industry-by-year fixed effects, instead of firm and year fixed effects. The use of industry-by-year fixed effects enables me to control for concurrent macroeconomic or regulatory events, allowing an exploration of the variation in *Exposure_Other* and *Exposure_En* across firms within the same industry and year. The result remains consistent. Finally, prior literature argues that using log or log-like transformation as the dependent variable may introduce bias (Cohn et al. 2022). They suggest using fixed effects poisson regression instead. I further address this concern by re-estimating the main specification using a fixed-effects Poisson model. As shown in panel B of **Table 14**, the results remain largely similar and consistent.

7 Conclusion

The recent accounting literature on patent disclosure has grown substantially (Glaeser et al. 2023). Yet, our understanding of how the search costs of patent information affect corporate innovation remains limited. This study provides novel empirical evidence on its effects by exploiting the adoption of the Cooperative Patent Classification (CPC) as a natural experiment that lowers

search costs for relevant patent information. My finding suggests that a one standard deviation increase in exposure to non-English European patent disclosures is associated with an approximately 7.66% increase in innovation activity among affected U.S. firms following the adoption of CPC. These gains are driven by improvements in the efficiency of knowledge recombination, as firms generate more innovation activity per unit of cumulative R&D, rely on more closely related prior knowledge in developing innovations, and increase their use of European patent information.

The positive impact of the adoption of CPC on corporate innovation is most pronounced among small firms and those without significant subsidiaries in Germany or France prior to its implementation. This finding suggests that the CPC democratizes the search for relevant European technical knowledge, benefiting firms that previously faced higher search barriers. However, the overall increase in innovation activity is partially offset by European import competition, indicating that the adoption of CPC is also a double-edged sword.

Further analysis using the private firm sample suggests that the results can extend to private firm, as private firms with greater exposure to European patents also experience innovation gains following the CPC adoption. Finally, aggregate analysis at the technology-field level indicates that the effects of the CPC adoption extend beyond individual firms and contribute to broader increases in overall innovation.

This paper contributes to the literature in several ways. First, it shows that the effectiveness of patent disclosure depends not only on whether information is made public, but also on how easily it can be searched. By focusing on the role of searchability, the paper highlights an important but underexplored dimension of the patent system (Glaeser et al. 2023). Second, it extends the literature on the real effects of information dissemination by examining patent information dissemination rather than financial disclosure dissemination. Finally, its findings contribute to the broader literature on patent system by demonstrating that information architecture—specifically, how patent information is classified—plays a central role in shaping innovation outcomes.

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Figure 1: Search Patent using CPC

By using CPC codes in Google Patents, one can easily find over one million patents related to “MOTOR VEHICLES; TRAILERS,” represented by the classification code B62D in the CPC system.

The screenshot shows the Google Patents search interface. The search bar contains the CPC code "B62D". The results show "More than 100,000 results". The top result is titled "Motor control apparatus" with the patent number "EP US CN JP • EP3608203B1" and the inventor "Naoki Shoji • Jtekt Corporation". The abstract describes a motor control apparatus for steering angle control, including a manual steering command value generation unit (41) and a steering torque (T d).

One can further narrow the search for relevant patents by adding more specific CPC classification codes, allowing users to identify patents that are most closely aligned with their research interests.

The screenshot shows the Google Patents search interface with the search bar containing the CPC codes "B62D H03K H02G". The results show "About 1 results". The top result is titled "Motorhome body device" with the patent number "EP DE • EP4461626A1" and the inventor "Karl-Heinz Schuler • Scolotech GmbH". The abstract describes a motorhome body device with at least two side wall modules (12, 14), a roof module (16), and a rear module (18).

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Fig. 1

Figure 3: The Relevance of non-English Written European Patent Disclosure

This figure plots the relevance of non-English written European patent disclosure. I present the prevalence of European patent disclosure across technology sections (not subclasses), because there are around 600 technology subclasses. The technology section's definition is following; A - Human necessities, B - Performing operations; transporting, C - Chemistry; metallurgy, D - Textiles; paper, E - Fixed constructions, F - Mechanical engineering; lighting; heating; weapons; blasting engines or pumps, G - Physics, H - Electricity, Y - New technology.

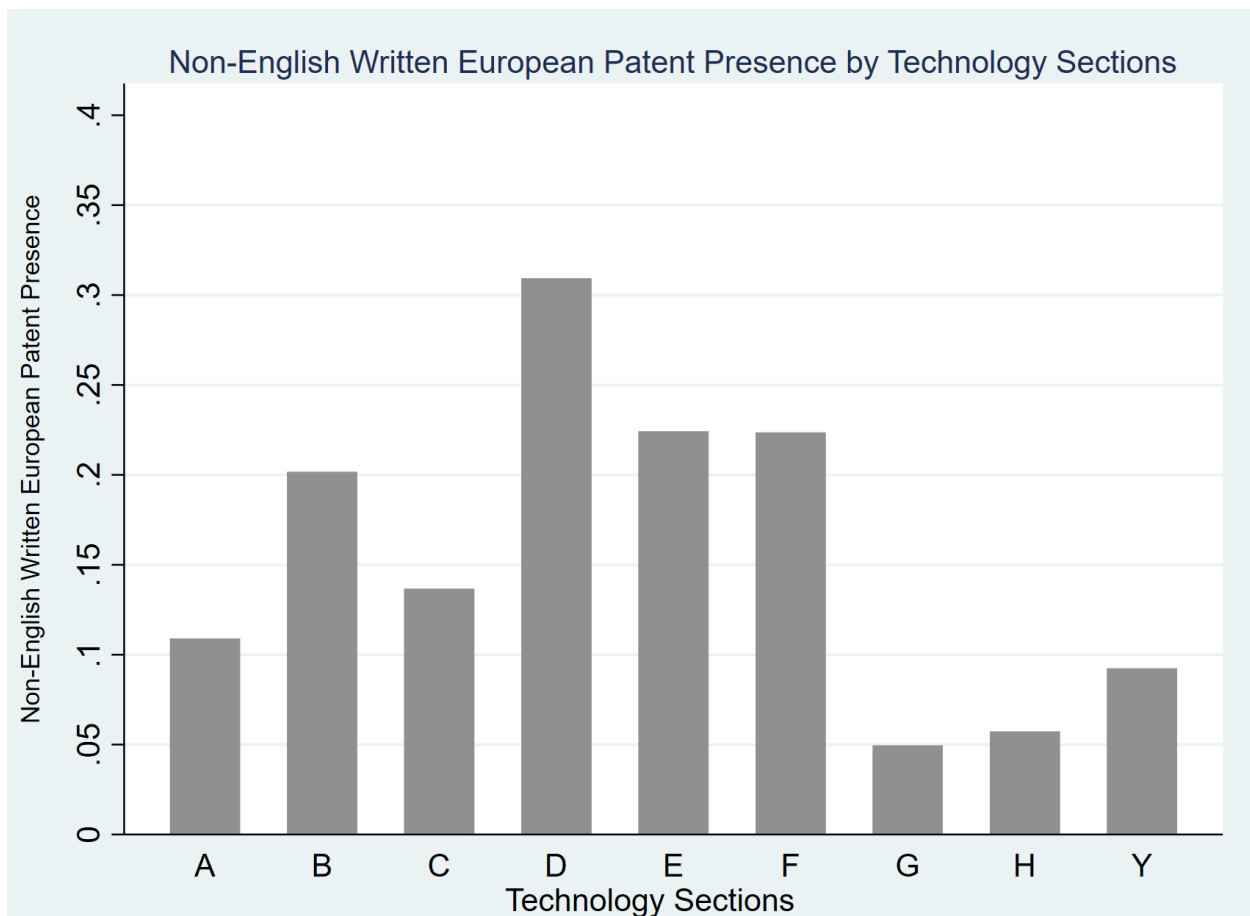


Figure 4: Pre-Trends in Innovation Activity

This figure plots the coefficient estimates on the time dummy variables of the dynamic regression that estimate the effect of the adoption of CPC on innovation activity. The dependent variable, *Forward Citation*, is the inverse hyperbolic transformation of the total number of US patent-generated forward citations a firm receives for all patents filed in year t , where forward citations are measured through 2024. The variable *Exposure_Other* are continuous variables which measure the firm's exposure to non-English written European patent disclosure. $Year_t$ is an indicator variable that equals 1 for year t . The X-axis represents the year relative to the adoption of CPC year. The Y-axis represents the coefficient estimates on the time indicator variables surrounding the adoption of CPC. Vertical bars represent 90% confidence intervals.

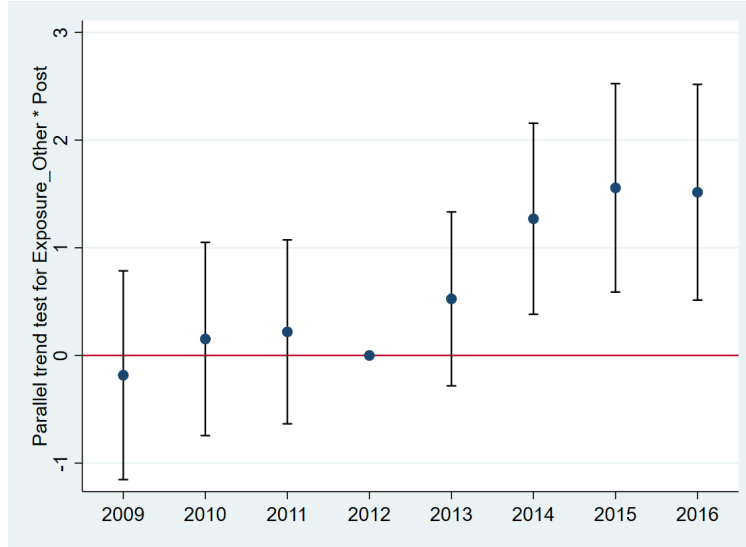
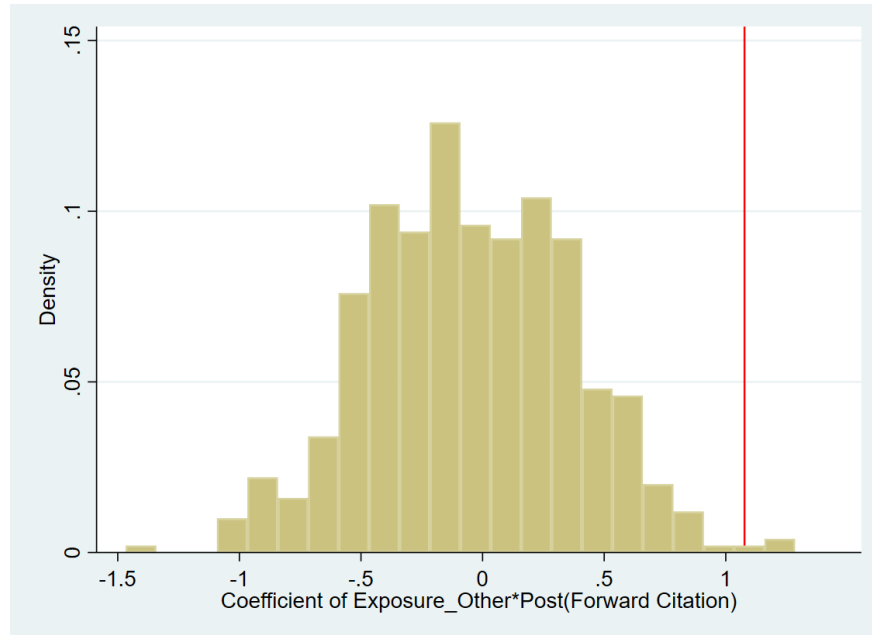


Figure 5: Placebo $Exposure_Other \times Post$ Coefficient

This figure displays the distribution of coefficients from 500 placebo regressions. For each experiment, I randomly shuffle $Exposure_Other$ and re-estimate the regression equation. It is clear to see that the distribution of estimates is centered around zero, contrasting starkly with the real estimated effects of β_1 represented by the solid red line



This figure displays the distribution of t-stats from 500 placebo regressions. For each experiment, I randomly shuffle $Exposure_Other$ and re-estimate the regression equation. It is clear to see that the distribution of t-stats is centered around zero.

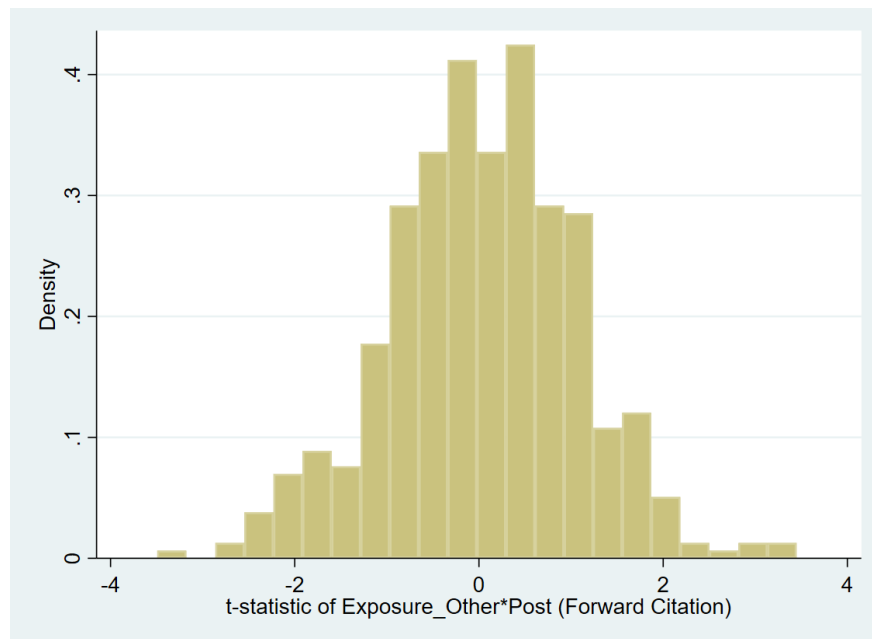


Table 1: Sample Selection

Criteria	Firm-year Observations
Public firm observations from 2009 to 2016	42,899
Require firms to have non-missing <i>Exposure_Other</i> and <i>Exposure_En</i>	16,050
Require firms to have non-missing primary dependent variables, control variables, and drop singleton observations	15,856

Notes: This table summarizes the sample selection criteria for our main analyses. Since I do not explicitly require non-missing values for alternative dependent variables or cross-sectional variables, the number of observations may be lower in some analyses depending on data availability.

Table 2: Summary Statistics

This table presents descriptive statistics for the key variables used in my analysis, including mean, standard deviation, and percentile values. Variable definitions are provided in the Appendix. All variables are presented in their raw form for the descriptive statistics.

Variable Name	Mean	SD	Min	P25	P50	P75	Max	N
Forward Citation	188.88	710.19	0.00	0.00	1.00	41.00	5255.00	15856
Forward Citation per Patent	4.01	8.56	0.00	0.00	0.50	4.20	56.67	15856
Scaled Forward Citation	0.01	0.05	0.00	0.00	0.00	0.00	0.36	15856
Patents	30.93	116.66	0.00	0.00	1.00	9.00	878.00	15856
Claims	542.24	1997.41	0.00	0.00	18.00	158.00	14909.00	15856
Patent Value	0.05	0.12	0.00	0.00	0.00	0.04	0.84	15856
Innovative Efficiency	0.60	2.06	0.00	0.00	0.00	0.25	15.72	15856
R&D Intensity	0.09	0.16	0.00	0.00	0.03	0.12	0.87	15365
Capex	0.04	0.04	0.00	0.01	0.03	0.05	0.23	15365
Net hire	-1.29	9.80	-64.00	-1.00	0.00	0.00	31.00	15856
EU	26.95	106.61	0.00	0.00	0.00	5.00	816.00	15856
EU Appl	26.36	104.23	0.00	0.00	0.00	5.00	796.00	15856
EU Other	4.48	17.55	0.00	0.00	0.00	1.00	131.00	15856
EU En	22.05	87.82	0.00	0.00	0.00	4.00	668.00	15856
Independent Variables								
Exposure_Other	0.11	0.07	0.02	0.05	0.10	0.15	0.31	15856
Exposure_En	0.20	0.13	0.04	0.12	0.17	0.25	0.56	15856
Exposure_Other Restrict	0.04	0.03	0.01	0.02	0.03	0.06	0.17	15856
Exposure_En Restrict	0.06	0.04	0.02	0.04	0.05	0.08	0.17	15856
High Exposure_Other	0.50	0.50	0.00	0.00	1.00	1.00	1.00	15856
High Exposure_En	0.50	0.50	0.00	0.00	1.00	1.00	1.00	15856
Controls								
Age	22.94	19.05	0.00	10.00	18.00	30.00	87.00	15856
Return on Asset	-0.06	0.29	-1.60	-0.06	0.03	0.07	0.29	15856
Leverage	0.19	0.20	0.00	0.00	0.15	0.30	0.95	15856
Firm Size	6.86	2.30	2.14	5.15	6.80	8.49	12.03	15856
Book to Market	0.54	0.47	-0.62	0.23	0.44	0.76	2.43	15856
Cash	0.27	0.26	0.00	0.07	0.18	0.40	0.97	15856
Number of Analyst	8.02	7.72	0.00	2.00	5.42	12.08	33.00	15856

Table 3: Innovation Activity

This table presents the results of the effect of CPC on innovation activity. The dependent variable, *Forward Citation*, is the IHS-transformed total number of US patent-generated forward citations a firm receives for all patents filed in year t , where forward citations are measured through 2024. The variable *Exposure_Other* and *Exposure_En* are continuous variables which separately measure the firm's exposure to non-English written European patent disclosure and English written European patent disclosure. *Post* is an indicator variable that equals 0 for the year 2009 - 2012 and 1 for the year 2013–2016. The t-statistics reported below the coefficient estimates in parentheses are computed based on standard errors clustered by firm.

	(1)	(2)	(3)	(4)
	<i>Forward Citation</i>	<i>Forward Citation</i>	<i>#Forward Citation/Patent</i>	<i>#Forward Citation/Patent</i>
Post × Exposure_Other	1.20*** (2.80)	1.16*** (2.74)	0.94*** (3.33)	0.88*** (3.16)
Post × Exposure_En	-0.25 (-1.00)	-0.33 (-1.36)	-0.06 (-0.38)	-0.06 (-0.38)
Return on Asset		-0.05 (-0.76)		-0.01 (-0.20)
Leverage		0.01 (0.10)		-0.11 (-1.13)
Firm Size		0.19*** (6.15)		0.08*** (3.81)
Book to Market		0.09* (1.70)		-0.01 (-0.36)
Cash		0.30** (2.26)		0.24** (2.59)
Number of Analyst		0.00 (0.47)		-0.00 (-1.02)
Age		0.11* (1.65)		0.07* (1.86)
Observations	15856	15856	15856	15856
Adjusted R^2	0.82	0.82	0.61	0.62
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

t statistics in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 4: The Real Effects of CPC

This table presents the results of the real effects of the adoption of CPC. The dependent variable, *R&D Intensity*, is the ratio of R&D expense to lagged total assets, where missing values are set to zero. The dependent variable, *Capex*, is the ratio of capital expenditure to lagged total assets, where missing values are set to zero. The dependent variable, *Net hire*, is the difference between the number of newly hired inventors and the number of leaving inventors. The variable *Exposure_Other* and *Exposure_En* are continuous variables which separately measure the firm's exposure to non-English written European patent disclosure and English written European patent disclosure. *Post* is an indicator variable that equals 0 for the year 2009 - 2012 and 1 for the year 2013 – 2016. The t-statistics reported below the coefficient estimates in parentheses are computed based on standard errors clustered by firm.

	(1) <i>R&D Intensity</i>	(2) <i>Capex</i>	(3) <i>Net Hire</i>
Post×Exposure_Other	0.09*** (3.12)	0.03*** (3.25)	5.67** (2.55)
Post × Exposure_En	-0.10*** (-3.35)	-0.00 (-0.17)	0.65 (0.53)
Return on Asset	-0.04*** (-3.85)	0.01*** (5.82)	0.12 (0.46)
Leverage	-0.01 (-1.20)	-0.01*** (-3.25)	-2.10*** (-3.16)
Firm Size	0.01*** (2.79)	0.01*** (8.66)	-0.03 (-0.18)
Book to Market	-0.01*** (-2.97)	-0.00*** (-3.11)	-0.84*** (-3.53)
Cash	0.02* (1.78)	-0.03*** (-7.50)	-0.38 (-0.68)
Age	-0.00*** (-2.72)	0.00 (0.82)	0.18 (0.25)
Number of Analyst	-0.00*** (-3.06)	0.00 (1.09)	-0.06 (-1.16)
Observations	15365	15365	15856
Adjusted R^2	0.83	0.67	0.60
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

t statistics in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 5: Efficient Knowledge Recombination

This table presents the result of the effect of CPC on innovative efficiency. The dependent variable, *Innovative Efficiency*, is the IHS-transformed ratio of the total number of US patent-generated forward citations a firm receives for all patents filed in a year t scaled by the R&D capital, where forward citations are measured through 2024. R&D capital is defined as the cumulative R&D expenses over the last five years, with a depreciation rate of 20%. The variable *Exposure_Other* and *Exposure_En* are continuous variables which separately measure the firm's exposure to non-English written European patent disclosure and English written European patent disclosure. *Post* is an indicator variable that equals 0 for the year 2009 - 2012 and 1 for the year 2013 – 2016. The t-statistics reported below the coefficient estimates in parentheses are computed based on standard errors clustered by firm.

	(1)	(2)
	<i>Innovation Efficiency</i>	<i>Innovation Efficiency</i>
Post×Exposure_Other	0.65*** (4.52)	0.58*** (4.08)
Post × Exposure_En	-0.02 (-0.28)	0.02 (0.22)
Return on Asset		0.02 (0.76)
Leverage		-0.10** (-1.97)
Firm Size		0.01 (0.52)
Book to Market		-0.05*** (-2.70)
Cash		0.23*** (4.00)
Age		-0.04*** (-2.91)
Number of Analyst		-0.01*** (-3.40)
Observations	15856	15856
Adjusted R^2	0.63	0.63
Firm FE	Yes	Yes
Year FE	Yes	Yes

t statistics in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 6: Focused Knowledge Recombination

This table presents the result of the effect of CPC on the technological distance between the U.S. and European knowledge cited by new patents and firms' existing knowledge base, using a patent–firm-level sample. The dependent variable, *Technological Distance*, captures how closely the US and European patents cited in a firm's patent application align with the firm's existing lines of technology—a smaller value indicates greater closeness. The variable *Exposure_Other* and *Exposure_En* are continuous variables which separately measure the firm's exposure to non-English written European patent disclosure and English written European patent disclosure. *Post* is an indicator variable that equals 0 for the year 2009 - 2012 and 1 for the year 2013 – 2016. Column (1) presents the baseline regression. Column (2) adds a vector of control variables. In both Columns, I additionally account for technology- and time-specific trends by incorporating CPC Subclass× filing year. The t-statistics reported below the coefficient estimates in parentheses are computed based on standard errors clustered by firm.

	(1)	(2)
	<i>Technological Distance</i>	<i>Technological Distance</i>
Post × Exposure_Other	-0.13** (-2.13)	-0.13** (-2.28)
Post × Exposure_En	-0.03 (-0.53)	-0.02 (0.34)
Return on Asset		-0.03* (1.89)
Leverage		0.03* (1.94)
Firm Size		0.01*** (2.81)
Book to Market		0.00 (0.76)
Cash		-0.00 (-0.22)
Age		0.01 (0.99)
Number of Analyst		0.00* (1.66)
Observations	588908	588908
Adjusted R^2	0.64	0.64
Firm FE	Yes	Yes
CPC Subclass-File Year FE	Yes	Yes

t statistics in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 7: Knowledge Flows Between U.S. and Europe

This table presents the result of the effect of CPC on knowledge flow. The dependent variable *EU* represents the IHS-transformed total number of European patents cited by either patent applicants or examiners in patent applications filed by a firm in year t . *EU Appl* captures the IHS-transformed total number of European patents cited specifically by applicants. *EU En* indicates the IHS-transformed total number of English-language European patents cited by either party, all based on patent applications filed by a firm in year t , while *EU Other* refers to the IHS-transformed total number of non-English European patents cited by either applicants or examiners. The variable *Exposure_Other* and *Exposure_En* are continuous variables which separately measure the firm's exposure to non-English written European patent disclosure and English written European patent disclosure. *Post* is an indicator variable that equals 0 for the year 2009 - 2012 and 1 for the year 2013 - 2016. The t-statistics reported below the coefficient estimates in parentheses are computed based on standard errors clustered by firm.

	(1)	(2)	(3)	(4)
	<i>EU</i>	<i>EU Appl</i>	<i>EU En</i>	<i>EU Other</i>
Post×Exposure_Other	0.61** (2.02)	0.57* (1.90)	0.45 (1.57)	0.82*** (3.65)
Post × Exposure_En	-0.22 (-1.17)	-0.22 (-1.16)	-0.20 (-1.06)	-0.04 (-0.30)
Return on Asset	-0.08 (-1.57)	-0.08 (-1.63)	-0.08 (-1.63)	-0.03 (-1.21)
Leverage	0.14 (1.40)	0.14 (1.40)	0.14 (1.48)	0.03 (0.47)
Firm Size	0.10*** (4.69)	0.10*** (4.70)	0.10*** (4.83)	0.03** (2.22)
Book to Market	0.05 (1.61)	0.05 (1.57)	0.05* (1.84)	0.00 (0.25)
Cash	0.17* (1.89)	0.16* (1.82)	0.19** (2.27)	0.01 (0.22)
Age	0.08 (1.39)	0.08 (1.38)	0.08 (1.39)	0.04 (0.84)
Number of Analyst	0.01 (1.41)	0.01 (1.47)	0.01 (1.38)	-0.00 (-0.03)
Observations	15856	15856	15856	15856
Adjusted R^2	0.84	0.84	0.84	0.82
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes

t statistics in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 8: Firms' ex ante information disadvantages

This table presents results on whether the effect of CPC adoption on innovation output varies by firms' ex ante information disadvantages. The dependent variable, *Forward Citation*, is the IHS-transformed total number of US patent-generated forward citations a firm receives for all patents filed in year t , measured through 2024. *No French and Germany Presence* equals to one if a firm had no significant subsidiaries in France or Germany in 2012 (pre-CPC adoption). *Small Firm* equals to one if the firm is below the median asset size in 2012. The variable *Exposure_Other* and *Exposure_En* are continuous variables which separately measure the firm's exposure to non-English written European patent disclosure and English written European patent disclosure. *Post* is an indicator for post-CPC adoption. Panel A examines firms without significant subsidiaries in France/Germany. Panel B examines small firms. t-statistics clustered by firm in parentheses.

Panel A. No France/Germany Presence

	(1)	(2)
	<i>Forward Citation</i>	<i>Forward Citation</i>
Post $\times$ Exposure_Other $\times$ No France/Germany Presence	2.46*** (2.65)	2.36** (2.53)
Post $\times$ Exposure_En $\times$ No France/Germany Presence	-0.97* (-1.69)	-0.90 (-1.57)
Post $\times$ Exposure_Other	-0.34 (-0.43)	-0.32 (-0.41)
Post $\times$ Exposure_En	0.38 (0.77)	0.23 (0.47)
Observations	14574	14574
Adjusted R^2	0.83	0.83
Controls	No	Yes
Firm FE	Yes	Yes
Year FE	Yes	Yes

t statistics in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Panel B. Small Firm

	(1)	(2)
	<i>Forward Citation</i>	<i>Forward Citation</i>
Post $\times$ Exposure_Other $\times$ Small	1.81** (2.10)	1.53* (1.79)
Post $\times$ Exposure_En $\times$ Small	-0.31 (-0.60)	-0.30 (-0.60)
Post $\times$ Exposure_Other	0.60 (1.01)	0.68 (1.15)
Post $\times$ Exposure_En	-0.18 (-0.44)	-0.28 (-0.71)
Observations	14574	14574
Adjusted R^2	0.83	0.83
Controls	No	Yes
Firm FE	Yes	Yes
Year FE	Yes	Yes

t statistics in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 9: Cross-sectional Tests: European Import Competition

This table presents results on whether the effect of CPC on innovation output is mitigated by import competition from Europe. The dependent variable, *Forward Citation*, is the IHS-transformed total number of US patent-generated forward citations a firm receives for all patents filed in year t , where forward citations are measured through 2024. *Low Competition* is an indicator variable equal to 1 if a firm faced below-median import competition from European firms in 2012, as measured by import penetration. The variable *Exposure_Other* and *Exposure_En* are continuous variables which separately measure the firm's exposure to non-English written European patent disclosure and English written European patent disclosure. *Post* is an indicator variable that equals 0 for the year 2009 - 2012 and 1 for the year 2013 – 2016. The t-statistics reported below the coefficient estimates in parentheses are computed based on standard errors clustered by firm.

	(1)	(2)
	<i>Forward Citation</i>	<i>Forward Citation</i>
Post $\times$ Exposure_Other $\times$ Low Competition	2.56*	2.52*
	(1.94)	(1.92)
Post $\times$ Exposure_En $\times$ Low Competition	-1.01	-0.80
	(-1.21)	(-0.97)
Post $\times$ Exposure_Other	1.65**	1.50*
	(2.01)	(1.82)
Post $\times$ Exposure_En	0.39	0.28
	(1.23)	(0.88)
Observations	9113	9113
Adjusted R^2	0.82	0.82
Controls	No	Yes
Firm FE	Yes	Yes
Year FE	Yes	Yes

t statistics in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 10: Private Firms

This table presents the result of the effect of CPC on private firms' innovation output. The dependent variable, *Patent*, is the IHS-transformed total number of granted patents that a private firm filed in a year. The dependent variable, *Claim*, is the IHS-transformed total number of patent claims a private firm has for all patents filed in a year. The dependent variable, *Forward Citation*, is the IHS-transformed total number of US patent-generated forward citations a private firm receives for all patents filed in a year. Forward citations are measured through 2024. The variable *Exposure_Other* and *Exposure_En* are continuous variables which separately measure the private firm's exposure to non-English written European patent disclosure and English written European patent disclosure. *Post* is an indicator variable that equals 0 for the year 2009 - 2012 and 1 for the year 2013 – 2016. The t-statistics reported below the coefficient estimates in parentheses are computed based on standard errors clustered by firm.

	(1)	(2)	(3)
	<i>Patent</i>	<i>Claim</i>	<i>Forward Citation</i>
Post $\times$ Exposure_Other	0.30*** (6.25)	0.97*** (8.02)	0.80*** (9.06)
Post $\times$ Exposure_En	0.06* (1.80)	0.21** (2.46)	0.33*** (5.50)
Observations	189304	189304	189304
Adjusted R^2	0.62	0.55	0.55
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

t statistics in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 11: Aggregate Effect

This table presents the result of CPC on CPC subclasses' innovation activity. The dependent variable, *Subclass Patent*, is the IHS-transformed number of granted patents filed in a year that are assigned to a CPC subclass. The dependent variable, *Subclass Claims*, is the IHS-transformed total number of patent claims across patents filed in a year that are assigned to a CPC subclass. The dependent variable, *Subclass Forward Citation*, is the IHS-transformed total number of US patent-generated forward citations received by patents filed in a year that are assigned to a CPC subclass. Forward citations are measured through 2024. The variable *Subclass Exposure_Other* and *Subclass Exposure_En* are continuous variables which separately measure the CPC subclass's exposure to non-English written European patent disclosure and English written European patent disclosure. *Post* is an indicator variable that equals 0 for the year 2009 - 2012 and 1 for the year 2013 – 2016. The t-statistics reported below the coefficient estimates in parentheses are computed based on standard errors clustered by firm.

	(1) <i>Subclass Patent</i>	(2) <i>Subclass Claim</i>	(3) <i>Subclass Forward Citation</i>
Post × Subclass Exposure_Other	0.20** (2.17)	0.19* (1.77)	0.11 (0.64)
Post × Subclass Exposure_En	-0.18 (-1.36)	-0.16 (-1.13)	-0.29 (-1.31)
Observations	5139	5139	5139
Adjusted R^2	0.97	0.97	0.93
CPC Subclass FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

t statistics in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 12: Quarterly Analysis

This table presents the results of the effect of CPC on innovation activity. The dependent variable, *Forward Citation*, is the IHS-transformed total number of US patent-generated forward citations a firm receives for all patents filed in quarter t , where forward citations are measured through 2024. The variable *Exposure_Other* and *Exposure_En* are continuous variables which separately measure the firm's exposure to non-English written European patent disclosure and English written European patent disclosure. *Post* is an indicator variable that equals 0 for the quarter before 2013Q1 and 1 for the year quarter after 2013Q1. All control variables are measured at the quarterly level. The t-statistics reported below the coefficient estimates in parentheses are computed based on standard errors clustered by firm.

	(1)	(2)
	<i>Forward Citation</i>	<i>Forward Citation</i>
Post $\times$ Exposure_Other	2.58*** (6.14)	2.41*** (5.86)
Post $\times$ Exposure_En	0.07 (0.31)	0.01 (0.04)
Return on Asset		0.20* (1.79)
Leverage		-0.08 (-0.83)
Firm Size		-0.12*** (-5.56)
Book to Market		-0.03 (-0.89)
Cash		0.62*** (6.01)
Number of Analyst		0.05*** (11.76)
Age		0.04* (1.65)
Observations	86145	86145
Adjusted R^2	0.59	0.60
Firm FE	Yes	Yes
Year-Quarter FE	Yes	Yes

t statistics in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 13: Robustness Tests - Alternative Innovation Proxies

This table presents whether the observed effect of CPC on innovation activity is robust to alternative innovation proxies. The dependent variable is the untransformed *Forward Citation* in Column (1), the log-transformed *Forward Citation* in Column (2), the untransformed *Forward Citation* scaled by the total number of citations received by all patents filed in the same year and multiplied by 100, *Scaled Forward Citation*, in Column (3), the IHS-transformed total number of patent claims that a firm filed in a year, *Patent*, in Column (4), a dummy equal to one if the firm filed for any subsequently granted patent in Column (5), the IHS-transformed total number of patent claims a firm has for all patents filed in a year, *Claims*, in Column (6), and the IHS-transformed sum of the economic value of a firm's patents filed in a year scaled by total assets, as developed by Kogan et al. 2017, *Patent Value*, in Column (7). The variable *Exposure_Other* and *Exposure_En* are continuous variables which separately measure the firm's exposure to non-English written European patent disclosure and English written European patent disclosure. *Post* is an indicator variable that equals 0 for the year 2009 - 2012 and 1 for the year 2013 – 2016. The t-statistics reported below the coefficient estimates in parentheses are computed based on standard errors clustered by firm.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	<i>Raw Forward Citation</i>	<i>Ln(1+Raw Forward Citation)</i>	<i>Scaled Forward Citation</i>	<i>Patent</i>	$1\{\#Patent > 0\}$	<i>Claims</i>	<i>Patent Value</i>
Post × Exposure_Other	520.40*** (4.57)	1.10*** (2.96)	0.02*** (2.93)	0.48* (1.81)	0.21** (1.98)	1.07** (2.10)	0.06** (2.46)
Post × Exposure_En	61.79 (1.50)	-0.29 (-1.37)	-0.00 (-0.76)	-0.28* (-1.90)	-0.08 (-1.31)	-0.34 (-1.17)	-0.05** (-2.23)
Observations	15,855	15,855	15,855	15,855	15,855	15,855	15,855
Adjusted R^2	0.86	0.83	0.90	0.89	0.63	0.81	0.75
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes

t statistics in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table 14: Robustness Tests - Alternative Model Specification

This table presents results on whether the observed effect of CPC on innovation output is robust to alternative Exposure_Other and Exposure_En proxies, fixed effects structure, and Poisson regression. The dependent variable, *Raw Forward Citation*, is the total number of US patent-generated forward citations a private firm receives for all patents filed in a year. Forward citations are measured through 2024. The dependent variable, *Forward Citation*, is the IHS-transformed total number of US patent-generated forward citations a firm receives for all patents filed in a year. Forward citations are measured through 2024. The dependent variable, *Raw Patent*, is the total number of granted patents that a private firm filed in a year. The dependent variable, *Raw Clam*, is the total number of patent claims a private firm has for all patents filed in a year. The dependent variable, *Raw Forward Citation*, is total number of US patent-generated forward citations a firm receives for all patents filed in a year. Forward citations are measured through 2024. *Exposure_Other Restrict* and *Exposure_En Restrict* are continuous variables which separately measure the firm's exposure to non-English written European patent disclosure and English written European patent disclosure, where European patents are defined as those filed exclusively in Europe. *Exposure_Other* and *Exposure_En* are continuous variables which separately measure the firm's exposure to non-English written European patent disclosure and English written European patent disclosure. *High Exposure_Other* equals one if *Exposure_Other* is above its median, and zero otherwise. *High Exposure_En* equals one if *Exposure_En* is above its median, and zero otherwise. *Post* is an indicator variable that equals 0 for the year 2009 - 2012 and 1 for the year 2013 – 2016. Panel A examines alternative *Exposure_Other* and *Exposure_En* proxies and fixed effects structures. Panel B presents results from Poisson regressions. The t-statistics reported below the coefficient estimates in parentheses are computed based on standard errors clustered by firm.

Panel A. Alternative Exposure_Other and Exposure_En Proxies

	(1) <i>Forward Citation</i>	(2) <i>Forward Citation</i>	(3) <i>Forward Citation</i>
Exposure_Other Restrict	2.55*** (3.16)		
Exposure_En Restrict	-0.55 (-0.69)		
High Exposure_Other		0.13* (1.82)	
High Exposure_En		-0.06 (-0.90)	
Exposure_Other			1.50*** (2.62)
Exposure_En			-0.44 (-0.95)
Observations	15856	15856	15851
Adjusted R^2	0.82	0.82	0.82
Controls	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	No
Industry-Year FE	No	No	Yes

t statistics in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Panel B. Poisson Regression

	(1) <i>Patent</i>	(2) <i>Claim</i>	(3) <i>Forward Citation</i>
Post × Exposure_Other	1.01** (2.41)	1.09** (2.41)	0.09 (0.12)
Post × Exposure_En	-0.85*** (-2.61)	-0.70** (-2.19)	-1.04** (-2.07)
Observations	12417	12417	11868
Controls	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

t statistics in parentheses

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

A Appendix

Variable	Definition
Dependent Variables	
Forward Citation	The IHS-transformed total number of US patent-generated forward citations a firm receives for all patents filed in year t . Forward citations are measured through 2024.
Forward Citation/Patent	The IHS-transformed average number of US patent-generated forward citations per patent that a firm receives for all patents filed in year t . Forward citations are measured through 2024.
Scaled Forward Citation	The untransformed <i>Forward Citation</i> scaled by the total number of citations received by all patents filed in the same year, multiplied by 100.
Patent	The IHS-transformed total number of granted patents that a firm filed in year t .
$1\{\# \text{ Patent} > 0\}$	An indicator variable that equals 1 if the firm applies for at least 1 granted patent in year t .
Claim	The IHS-transformed total number of patent claims a firm has for all patents filed in year t .
Patent Value	The IHS-transformed sum of the economic value of a firm's patents filed in year t , estimated by the grant day market reaction based on the method of Kogan et al. 2017, divided by the total assets.

Variable	Definition
Innovative Efficiency	The IHS-transformed ratio of the total number of US patent-generated forward citations a firm receives for all patents filed in a year t scaled by the R&D capital. R&D capital is defined as the cumulative R&D expenses over the last five years, with a depreciation rate of 20%.
R&D Intensity	The ratio of R&D expense to lagged total assets. Missing values are set to zero.
Capex	The ratio of capital expenditure to lagged total assets. Missing values are set to zero.
Net hire	The difference between the number of newly hired inventors and the number of leaving inventors.
Technology Distance	The average technological distance between the U.S. and European patents cited in a firm's patent application and the firm's existing patent portfolio. A lower value indicates that the cited prior arts are more closely related to the firm's technological expertise. See Appendix B for details.
EU	The IHS-transformed total number of European patents cited by patent applicants or patent examiners for patent applications filed by a firm in year t .
EU Appl	The IHS-transformed total number of European patents cited by patent applicants for patent applications filed by a firm in year t .
EU Other	The IHS-transformed total number of non-English European patents cited by patent applicants or patent examiners for patent applications filed by a firm in year t .
EU En	The IHS-transformed total number of English-written European patents cited by patent applicants or patent examiners for patent applications filed by a firm in year t .

Variable	Definition
Subclass Patent	The IHS-transformed total number of granted patents filed in year t that are assigned to a CPC subclass.
Subclass Claim	The IHS-transformed total number of patent claims across patents filed in year t that are assigned to a CPC subclass.
Subclass Forward Citation	The IHS-transformed total number of US patent-generated forward citations received by patents filed in year t that are assigned to a CPC subclass. Forward citations are measured through 2024.
Independent Variables	
Exposure_Other	A firm's exposure to non-English European patent disclosure, where detailed calculation procedure is in section 4.2
Exposure_En	A firm's exposure to English European patent disclosure, where detailed calculation procedure is in section 4.2.
Exposure_Other Restrict	A firm's exposure to non-English European patent disclosure, where I define European patents as those filed exclusively in Europe and follow detailed calculation procedure is in section 4.2
Exposure_En Restrict	A firm's exposure to English European patent disclosure, where I define European patents as those filed exclusively in Europe and follow detailed calculation procedure is in section 4.2
High Exposure_Other	Indicator variable equals to one if a firm's exposure to non-English European patent disclosure is above the median of Exposure_Other.
High Exposure_En	Indicator variable equals to one if a firm's exposure to English European patent disclosure is above the median of Exposure_En.

Variable	Definition
Subclass Exposure_Other	Technology subclass exposure to non-English European patent disclosure.
Subclass Exposure_En	Technology subclass exposure to English European patent disclosure.
Controls	
Age	The number of years since IPO from CRSP.
Return on Asset	Operating income before extraordinary items divided by total assets (IB/AT).
Leverage	Total liabilities divided by total assets (DLTT+DLC)/AT.
Firm Size	The natural log of one plus market capitalization.
Book to Market	Book value of equity divided by market capitalization (CEQ+TXDB)/(PRCC_F×CSHO).
Cash	Cash holdings divided by total assets (CHE/AT).
Number of Analysts	The average number of analysts that follow this firm in year t .
Cross-Sectional Variables	
Small	Indicator variable equals to one if the firm is below the median asset size in 2012
No France/Germany Presence	Indicator variable equals to one if a firm had no significant subsidiaries in France or Germany in 2012
Low Competition	Indicator variable equals to 1 if a firm faced below-median import competition from European firms in 2012

B Appendix - Construction of Technological Distance

I follow the methodology in prior literature to construct a patent's technological distance, which captures how closely a given patent aligns with the owning firm's existing knowledge base, as reflected in its prior patent portfolio. (Zhang 2024; Akcigit et al. 2016) The construction of this measure involves two steps. First, I quantify the proximity between technology classes by analyzing their citation patterns. Here, the technology class is at the CPC subclass level. Specifically, the technological closeness between patent technology classes X and Y is calculated using the following expression:

$$d(T_X, T_Y) \equiv 1 - \frac{\#(T_X \cap T_Y)}{\#(T_X \cup T_Y)}$$

The numerator, $\#(T_X \cap T_Y)$, represents the number of patents that cite both technology class X and technology class Y . The denominator, $\#(T_X \cup T_Y)$, captures the number of patents that cite either class X or class Y . This symmetric measure is intuitive: among all patents that cite either technology class X or Y , a larger number of patents citing both classes suggests a higher degree of proximity between the two classes in the technological knowledge space. The closer this measure, $d(T_X, T_Y)$, to 0, the more proximate the technology class X is to technology class Y .

Next, I calculate the technological distance between a patent p and the owning firm's existing patent portfolio, $d_\iota(p, P_f)$, can be calculated as follows.

$$d_\iota(p, P_f) \equiv \left[\frac{1}{\|P_f\|} \sum_{p' \in P_f} d(T_p, T_{p'})^\iota \right]^{\frac{1}{\iota}}$$

Specifically, to calculate the technological distance of the cited patent p from the owning firm's existing patent portfolio P_f (i.e., portfolio of all the patents that had been invented prior to citing patent p), I figure out the distance between technology class of patent p and that of every other patent p' in the patent portfolio P_f . Next, I aggregate these individual technological distances into a single master variable. Here, $\|P_f\|$ denotes the number of patents in the firm's patent portfolio,

and ι is set to $2/3$ following the existing literature (but my results are robust to different values of ι , e.g., $\iota = 1/3$ or $\iota = 1$). The larger this measure, the further away the focal patent is from the owning firm's knowledge space.

To measure how closely the prior patents cited in a firm's patent application relate to the firm's knowledge space, I calculate the average technological distance of the US and European patents cited in a firm's patent application from the firm's existing patent portfolio, which I refer to as *Technology Distance*. A smaller value of *Technology Distance* indicates that the cited US and European patents in a firm's patent application are more closely related to the firm's knowledge space.